# Language Technology

## Chapter 12: Words, Parts of Speech, and Morphology

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# The Parts of Speech

Parts of speech (POS) are classes that correspond to lexical (or word) categories.

Plato distinguished between the verb and the noun.

After him, the inventory of word categories continued to evolve and expand until Dionysius Thrax formalized and stabilized it.

Aelius Donatus later popularized the list of the eight classical parts of speech: noun, pronoun, verb, participle, conjunction, adverb, preposition, and interjection.

Grammarians have adopted these categories for most European languages, even though the classification remains somewhat arbitrary.

Parts of speech fall into two main classes: the open class and the closed class.

# Parts of Speech: Open-Class Words

| **POS** | **English** | **French** | **German** |
|---|---|---|---|
| Nouns | *name, Frank* | *nom, François* | *Name, Franz* |
| Adjectives | *big, good* | *grand, bon* | *groß, gut* |
| Verbs | *to swim* | *nager* | *schwimmen* |
| Adverbs | *rather, very, only* | *plutôt, très, uniquement* | *fast, nur, sehr, endlich* |

# Parts of Speech: Closed-Class Words

| **POS** | **English** | **French** | **German** |
|---|---|---|---|
| Determiners | *the, several, my* | *le, plusieurs, mon* | *der, mehrere, mein* |
| Pronouns | *he, she, it* | *il, elle, lui* | *er, sie, ihm* |
| Prepositions | *to, of* | *vers, de* | *nach, von* |
| Conjunctions | *and, or* | *et, ou* | *und, oder* |
| Auxiliaries and modals | *be, have, will, would* | *être, avoir, pouvoir* | *sein, haben, können* |

# Annotation with Parts of Speech

Sentence:

*That round table might collapse*

Annotation:

| **Word** | **Part of speech** | **POS tag** |
|---|---|---|
| *that* | Determiner | DET |
| *round* | Adjective | ADJ |
| *table* | Noun | NOUN |
| *might* | Modal verb | AUX |
| *collapse* | Verb | VERB |

Automatic annotation relies on predefined POS tagsets, such as the Penn Treebank tagset for English.

# Training Sets: The CoNLL Format

The CoNLL format is a tabular format used to distribute annotated texts. It was originally created for the shared tasks of the Conference on Natural Language Learning.

The precise annotation scheme has varied over the years; we use CoNLL-U, the most recent version.

Annotation of the Spanish sentence:

> *La reestructuración de los otros bancos checos se está acompañando por la reducción del personal*
> *'The restructuring of the other Czech banks is being accompanied by a reduction in staff'*

# Example of Annotation (CoNLL-U)

*La reestructuración de los otros bancos checos se está acompañando por la reducción del personal*

<table>
<tr><th>ID</th><th>FORM</th><th>LEMMA</th><th>UPOS</th><th>FEATS</th></tr>
<tr><td>1</td><td>La</td><td>el</td><td>DET</td><td>Definite=Def|Gender=Fem|Number=Sing|PronType=Art</td></tr>
<tr><td>2</td><td>reestructuración</td><td>reestructuración</td><td>NOUN</td><td>Gender=Fem|Number=Sing</td></tr>
<tr><td>3</td><td>de</td><td>de</td><td>ADP</td><td>AdpType=Prep</td></tr>
<tr><td>4</td><td>los</td><td>el</td><td>DET</td><td>Definite=Def|Gender=Masc|Number=Plur|PronType=Art</td></tr>
<tr><td>5</td><td>otros</td><td>otro</td><td>DET</td><td>Gender=Masc|Number=Plur|PronType=Ind</td></tr>
<tr><td>6</td><td>bancos</td><td>banco</td><td>NOUN</td><td>Gender=Masc|Number=Plur</td></tr>
<tr><td>7</td><td>checos</td><td>checo</td><td>ADJ</td><td>Gender=Masc|Number=Plur</td></tr>
<tr><td>8</td><td>se</td><td>se</td><td>PRON</td><td>Case=Acc|Person=3|PrepCase=Npr|PronType=Prs|Reflex=Yes</td></tr>
<tr><td>9</td><td>está</td><td>estar</td><td>AUX</td><td>Mood=Ind|Number=Sing|Person=3|Tense=Pres|VerbForm=Fin</td></tr>
<tr><td>10</td><td>acompañando</td><td>acompañar</td><td>VERB</td><td>VerbForm=Ger</td></tr>
<tr><td>11</td><td>por</td><td>por</td><td>ADP</td><td>AdpType=Prep</td></tr>
<tr><td>12</td><td>la</td><td>el</td><td>DET</td><td>Definite=Def|Gender=Fem|Number=Sing|PronType=Art</td></tr>
<tr><td>13</td><td>reducción</td><td>reducción</td><td>NOUN</td><td>Gender=Fem|Number=Sing</td></tr>
<tr><td>14</td><td>del</td><td>del</td><td>ADP</td><td>AdpType=Preppron</td></tr>
<tr><td>15</td><td>personal</td><td>personal</td><td>NOUN</td><td>Gender=Masc|Number=Sing</td></tr>
<tr><td>16</td><td>.</td><td>.</td><td>PUNCT</td><td>PunctType=Peri</td></tr>
</table>

# Part-of-Speech Annotation (CoNLL 2000)

Annotation of: *He reckons the current account deficit will narrow to only # 1.8 billion in September.* (The last column is set aside for now.)

| | | |
|---|---|---|
| He | PRP | B-NP |
| reckons | VBZ | B-VP |
| the | DT | B-NP |
| current | JJ | I-NP |
| account | NN | I-NP |
| deficit | NN | I-NP |
| will | MD | B-VP |
| narrow | VB | I-VP |
| to | TO | B-PP |
| only | RB | B-NP |
| # | # | I-NP |
| 1.8 | CD | I-NP |
| billion | CD | I-NP |
| in | IN | B-PP |
| September | NNP | B-NP |
| . | . | O |

# Part-of-Speech Annotation (Universal Dependencies)

Annotation of: *Do museum labels have an impact on how people look at artworks?*

| ID | FORM | LEMMA | UPOS | XPOS | FEATS |
|---|---|---|---|---|---|
| 1 | Do | do | AUX | VBP | Mood=Ind\|Number=Plur\|Person=3\|Tense=Pres\|VerbForm=Fin |
| 2 | museum | museum | NOUN | NN | Number=Sing |
| 3 | labels | label | NOUN | NNS | Number=Plur |
| 4 | have | have | VERB | VB | VerbForm=Inf |
| 5 | an | a | DET | DT | Definite=Ind\|PronType=Art |
| 6 | impact | impact | NOUN | NN | Number=Sing |
| 7 | on | on | ADP | IN | _ |
| 8 | how | how | SCONJ | WRB | PronType=Rel |
| 9 | people | person | NOUN | NNS | Number=Plur |
| 10 | look | look | VERB | VBP | Mood=Ind\|Number=Plur\|Person=3\|Tense=Pres\|VerbForm=Fin |
| 11 | at | at | ADP | IN | _ |
| 12 | artworks | artwork | NOUN | NNS | Number=Plur |
| 13 | ? | ? | PUNCT | . | _ |

# Part-of-Speech Annotation (Universal Dependencies)

Annotation of: *Genom skattereformen införs individuell beskattning (särbeskattning) av arbetsinkomster.*

| ID | FORM | LEMMA | UPOS | XPOS | FEATS |
|---|---|---|---|---|---|
| 1 | Genom | genom | ADP | PP | _ |
| 2 | skattereformen | skattereform | NOUN | NN\|UTR\|SIN\|DEF\|NOM | Case=Nom\|Definite=Def\|Gender=Com\|Number=Sing |
| 3 | införs | införa | VERB | VB\|PRS\|SFO | Mood=Ind\|Tense=Pres\|VerbForm=Fin\|Voice=Pass |
| 4 | individuell | individuell | ADJ | JJ\|POS\|UTR\|SIN\|IND\|NOM | Case=Nom\|Definite=Ind\|Degree=Pos\|Gender=Com\|Number=Sing |
| 5 | beskattning | beskattning | NOUN | NN\|UTR\|SIN\|IND\|NOM | Case=Nom\|Definite=Ind\|Gender=Com\|Number=Sing |
| 6 | ( | ( | PUNCT | PAD | _ |
| 7 | särbeskattning | särbeskattning | NOUN | NN\|UTR\|SIN\|IND\|NOM | Case=Nom\|Definite=Ind\|Gender=Com\|Number=Sing |
| 8 | ) | ) | PUNCT | PAD | _ |
| 9 | av | av | ADP | PP | _ |
| 10 | arbetsinkomster | arbetsinkomst | NOUN | NN\|UTR\|PLU\|IND\|NOM | Case=Nom\|Definite=Ind\|Gender=Com\|Number=Plur |
| 11 | . | . | PUNCT | MAD | _ |

# Ambiguity

| Word | Possible tags | Example of use |
|---|---|---|
| *that* | Subordinating conjunction | *That he can swim is good* |
| | Determiner | *That white table* |
| | Adverb | *It is not that easy* |
| | Pronoun | *That is the table* |
| | Relative pronoun | *The table that collapsed* |
| *round* | Verb | *Round up the usual suspects* |
| | Preposition | *Turn round the corner* |
| | Noun | *A big round* |
| | Adjective | *A round box* |
| | Adverb | *He went round* |
| *table* | Noun | *That white table* |
| | Verb | *I table that* |
| *might* | Noun | *The might of the wind* |
| | Modal verb | *She might come* |
| *collapse* | Noun | *The collapse of the empire* |
| | Verb | *The empire can collapse* |

# Part-of-Speech Ambiguity in Swedish

The word *som* in the *Norstedts svenska ordbok* (1999) has three entries:

1. *Om jag vore lika vacker som du, skulle jag vara lycklig.* (conjunction)
2. *Bilen som jag köpte i fjol.* (pronoun)
3. *Som jag har saknat dig.* (adverb)

The part-of-speech distinction can be significant:

**Swedish.** Compare the pronunciation of *vaken* (adjective) in *Han är aldrig vaken innan klockan sju* with *vaken* (noun) in *Vi fiskade i vaken i sjön*.

**English.** Compare *object* in *I object to violence* (verb) with *I could see an object* (noun).

# Grammatical Features

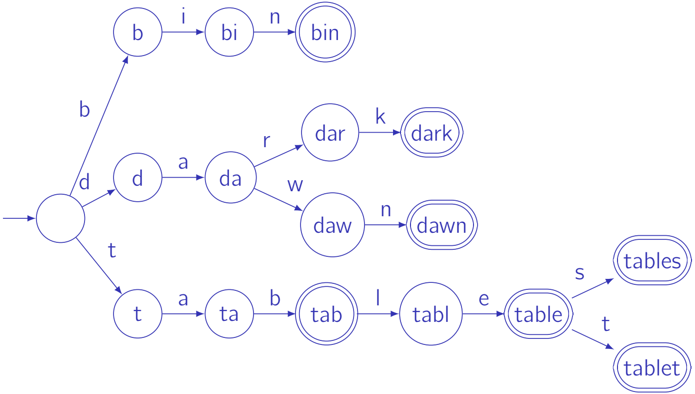
| Main parts of speech | Features (subcategories) |
|---|---|
| Adjective, noun, pronoun | Regular, base, comparative, superlative, interrogative, person, number, case |
| Adverb | Regular, base, comparative, superlative, interrogative |
| Article, determiner, preposition | Person, case, number |
| Verb | Tense, voice, mood, person, number, case |

# Lexicons: An Excerpt from the Oxford Advanced Learner's Dictionary

| Word | Pronunciation | Syntactic tag | Syllable count or verb pattern |
|---|---|---|---|
| a | @ | S-* | 1 |
| a | EI | Ki$ | 1 |
| a fortiori | eI ,fOtI'OraI | Pu$ | 5 |
| a posteriori | eI ,p0sterI'OraI | OA$,Pu$ | 6 |
| a priori | eI ,praI'OraI | OA$, Pu$ | 4 |
| a's | Eiz | Kj$ | 1 |
| ab initio | &b I'nISI@U | Pu$ | 5 |
| abaci | '&b@saI | Kj$ | 3 |
| aback | @'b&k | Pu% | 2 |
| abacus | '&b@k@s | K7% | 3 |
| abacuses | '&b@k@sIz | Kj% | 4 |
| abaft | @'bAft | Pu$,T-$ | 2 |
| abandon | @'b&nd@n | H0%,L@% | 36A,14 |
| abandoned | @'b&nd@nd | Hc%,Hd%,OA% | 36A,14 |
| abandoning | @'b&nd@nIN | Hb% | 46A,14 |
| abandonment | @'b&nd@nm@nt | L@% | 4 |
| abandons | @'b&nd@nz | Ha% | 36A,14 |
| abase | @'beIs | H2% | 26B |
| abased | @'beIst | Hc%,Hd% | 26B |
| abasement | @'beIsm@nt | L@% | 3 |

Newer standard:
https://tei-c.org/release/doc/tei-p5-doc/en/html/DI.html

# Letter Trees




# Morphemes

| | Word | Morpheme decomposition |
|---|---|---|
| English | *disentangling* | *dis*+*en*+**tangle**+*ing* |
| | *rewritten* | *re*+**write**+*en* |
| French | *désembrouillé* | *dé*+*em*+**brouiller**+*é* |
| | *récrite* | *re*+**écrire**+*te* |
| German | *entwirrend* | *ent*+**wirren**+*end* |
| | *wiedergeschrieben* | *wieder*+*ge*+**schreiben**+*en* |

# Inflection

| | Plural of nouns | Morpheme decomposition |
|---|---|---|
| English | *hedgehogs* | *hedgehog+s* |
| | *churches* | *church+es* |
| | *sheep* | *sheep+∅* |
| French | *hérissons* | *hérisson+s* |
| | *chevaux* | *cheval+ux* |
| German | *Gründe* | *Grund+(¨)e* |
| | *Hände* | *Hand+(¨)e* |
| | *Igel* | *Igel+∅* |

# Derivation

Creation of a new word:

| | English | French | German |
|---|---|---|---|
| Prefixes | **fore***see*, | **pré***voir*, | **vorher***sehen*, |
| | **un***pleasant* | **dé***plaisant* | **un***angenehm* |
| Suffixes | *manage***able**, | *gér***able**, | *vorsicht***ich**, |
| | *rigor***ous** | *rigour***eux** | *streit***bar** |

# Morphological Processing

**Generation →**

| **English** | | **French** | | **German** | |
|---|---|---|---|---|---|
| *dog+s* | *dogs* | *chien+s* | *chiens* | *Hund+e* | *Hunde* |
| *work+ing* | *working* | *travailler+ant* | *travaillant* | *arbeiten+end* | *arbeitend* |
| *un+do* | *undo* | *dé+faire* | *défaire* | | |

**← Parsing**

# Language Differences (Source: Xerox)

| Language | # stems | # inflected forms | | Lex. size (kb) |
|---|---|---|---|---|
| English | 55,000 | 240,000 | | 200–300 |
| French | 50,000 | 5,700,000 | | 200–300 |
| German | 50,000 | 350,000 | or | 450 |
| | | infinite | (compounding) | |
| Japanese | 130,000 | 200 | suffixes | 500 |
| | | 20,000,000 | word forms | 500 |
| Spanish | 40,000 | 3,000,000 | | 200–300 |

# Two-Level Morphology

Modern morphological parsers are based on the two-level model of Kimmo Koskenniemi (1983).

The model relates the surface form of a word (the form that appears in a text) to its lexical (or underlying) form (its sequence of morphemes):

| | |
|---|---|
| **Surface:** | disentangled |
| **Lexical (or underlying):** | dis+en+tangle+ed |

# Examples

| | | |
|---|---|---|
| **Generation**: Lexical to surface form $\rightarrow$ | | |
| English | *dis+en+tangle+ed* | *disentangled* |
| | *happy+er* | *happier* |
| | *move+ed* | *moved* |
| French | *dés+em+brouiller+é* | *désembrouillé* |
| | *dé+chanter+erons* | *déchanterons* |
| German | *ent+wirren+end* | *entwirrend* |
| | *wieder+ge+schreiben+en* | *wiedergeschrieben* |
| **Parsing**: $\leftarrow$ Surface to lexical form | | |

# Aligning the Two Forms

| | | | |
|---|---|---|---|
| English | dis+en+tangle+ed<br>$\updownarrow\updownarrow \ldots$<br>dis0en0tangl00ed | happy+er<br>$\updownarrow\updownarrow \ldots$<br>happi0er | move+ed<br>$\updownarrow\updownarrow \ldots$<br>mov00ed |
| French | dé+chanter+erons<br>$\updownarrow\updownarrow \ldots$<br>dé0chant000erons | cheval+ux<br>$\updownarrow\updownarrow \ldots$<br>cheva00ux | cheviller+é<br>$\updownarrow\updownarrow \ldots$<br>chevill000é |
| German | singen+st<br>$\updownarrow\updownarrow \ldots$<br>singe00st | Grund+¨e<br>$\updownarrow\updownarrow \ldots$<br>Gründ00e | Igel+∅<br>$\updownarrow\updownarrow \ldots$<br>Igel00 |

# Interpreting the Morphemes

Suffixes carry grammatical information: the French ending *erons*, for example, encodes verb + future + first person + plural.

Morphological parsers can therefore represent the lexical form as a concatenation of the stem and its grammatical features rather than the stem and the surface suffix.

The Xerox parser produces the following analyses for *disentangled* and *happier*:

```
disentangle+Verb+PastBoth+123SP
happy+Adj+Comp
```

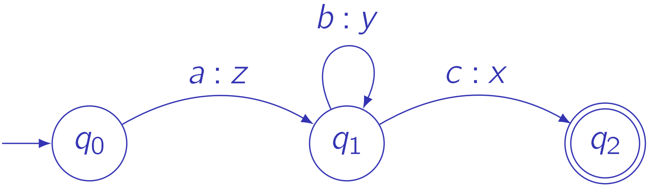
where `+Verb` marks a verb, `+PastBoth` indicates either past tense or past participle, and `+123SP` stands for any person, singular or plural; `+Adj` marks an adjective and `+Comp` a comparative.

# Aligning Morphemes and Features

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Lexical: | d | i | s | e | n | t | a | n | g | l | e | +Verb | +PastBoth | +123sp |
| Surface: | d | i | s | e | n | t | a | n | g | l | 0 | 0 | e | d |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Lexical: | h | a | p | p | y | +Adj | +Comp |
| Surface: | h | a | p | p | i | e | r |

# Transducers




The string `abbbc` is transduced into `zyyyx`.

# Mathematical Definition of a Finite-State Transducer

1. $Q$ is a finite set of states.
2. $\Sigma$ is a finite set of symbol pairs $i : o$, where $i$ belongs to the input alphabet and $o$ to the output alphabet. Both alphabets may include the empty symbol $\varepsilon$.
3. $q_0$ is the start state ($q_0 \in Q$).
4. $F$ is the set of final states ($F \subseteq Q$).
5. $\delta$ is the transition function $Q \times \Sigma \rightarrow Q$, where $\delta(q, i : o)$ returns the state reached from $q$ on the input–output pair $i : o$.

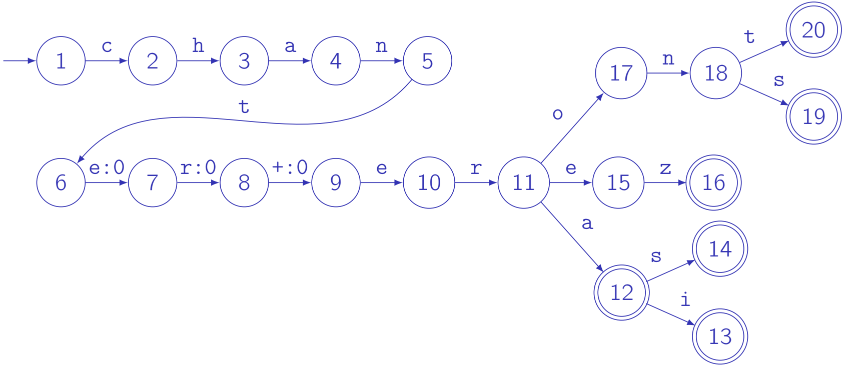
For the example transducer we have $Q = \{q_0, q_1, q_2\}$,
$\Sigma = \{a : z, b : y, c : x\}$,
$\delta = \{\delta(q_0, a : z) = q_1, \delta(q_1, b : y) = q_1, \delta(q_1, c : x) = q_2\}$, and $F = \{q_2\}$.

# French Verb Transducers for *chanter*

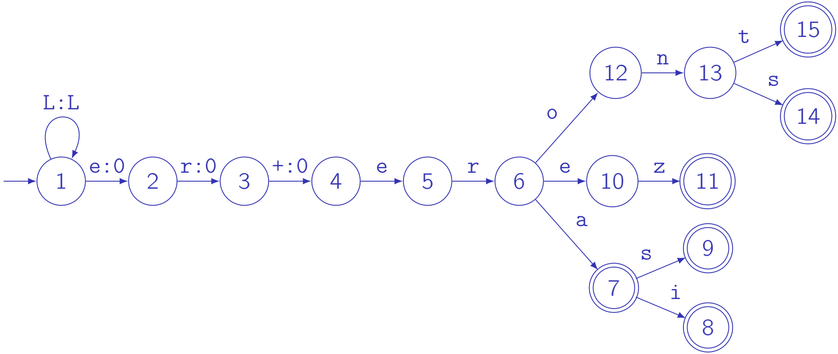
| **Number\Person** | **First** | **Second** | **Third** |
|---|---|---|---|
| singular | *chanterai* | *chanteras* | *chantera* |
| plural | *chanterons* | *chanterez* | *chanteront* |

| **Number\Pers.** | **First** | **Second** | **Third** |
|---|---|---|---|
| singular | `chanter+erai` | `chanter+eras` | `chanter+era` |
| | `chant000erai` | `chant000eras` | `chant000era` |
| plural | `chanter+erons` | `chanter+erez` | `chanter+eront` |
| | `chant000erons` | `chant000erez` | `chant000eront` |

# Transducer for *chanter*



# French Verb Transducers: Future, First Group




Such transducers can be implemented in Prolog or with OpenFst (https://www.openfst.org/).

# Romance Languages

| Language | Number\Person | First | Second | Third |
|---|---|---|---|---|
| Italian | | | | |
| | singular | *canterò* | *canterai* | *canterà* |
| | plural | *canteremo* | *canterete* | *canteranno* |
| Spanish | | | | |
| | singular | *cantaré* | *cantarás* | *cantará* |
| | plural | *cantaremos* | *cantaréis* | *cantarán* |
| Portuguese | | | | |
| | singular | *cantarei* | *cantarás* | *cantará* |
| | plural | *cantaremos* | *cantareis* | *cantarão* |

# Course Content: Optional Part

Starting in 2023, the remaining slides in this document are no longer part of the course.

You may nevertheless read them if you are curious.

# Ambiguity

In the future-tense transducer there is no ambiguity: each surface form corresponds to a unique lexical form and a unique final state.

This is not the case for the present tense:

> *(je) chante 'I sing'*
> *(il) chante 'he sings'*

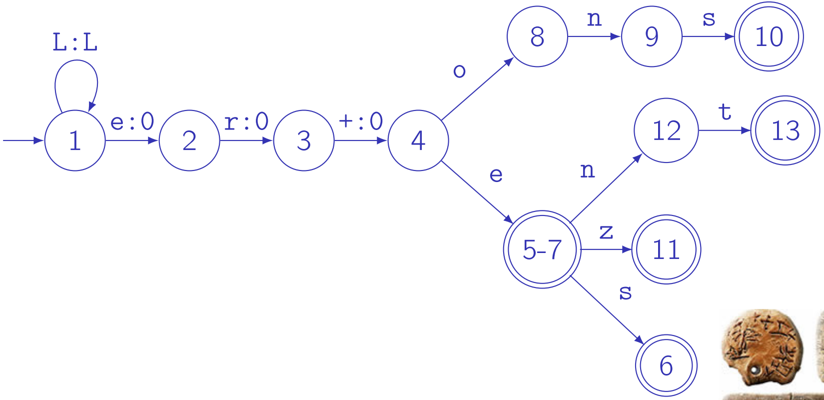
| **Number\Person** | **First** | **Second** | **Third** |
|---|---|---|---|
| singular | *chante* | *chantes* | *chante* |
| plural | *chantons* | *chantez* | *chantent* |

# Transducer Ambiguity

Final states 5 and 7 coincide.

A Prolog implementation is analogous to that of the future tense.

By backtracking, the transducer can return all final states and thereby expose the morphological ambiguity.



# Koskenniemi's Rules

Koskenniemi described morphology with declarative rules that refer to left and right contexts and employ the operators $\Rightarrow$, $\Leftarrow$, $\Leftrightarrow$, or $/ \Leftarrow$.

In English a lexical *y* may correspond to a surface *i*, as in *happier*.

This happens when *y* is preceded by a consonant and followed by *-er*, *-ed* or *-s*:

1. `y:i ⇐ C:C __ +:0 e:e r:r`
2. `y:i ⇐ C:C __ +:e s:s`
3. `y:i ⇐ C:C __ +:0 e:e d:d`

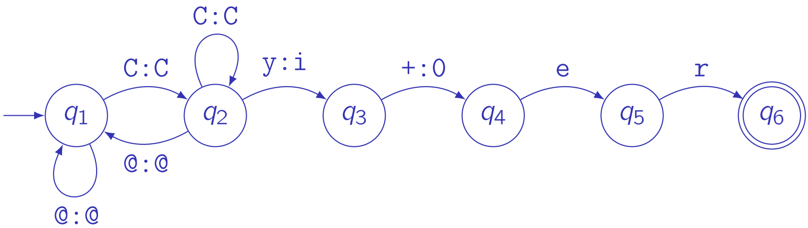
# Two-Level Rules

Lexical-to-surface transduction is expressed by rules of the following kinds:

| Rules | | | Description |
|---|---|---|---|
| `a:b` | $\Rightarrow$ | `lc __ rc` | `a` is realized as `b` **only** in the context `lc __ rc` |
| `a:b` | $\Leftarrow$ | `lc __ rc` | `a` is **always** realized as `b` in the context `lc __ rc` |
| `a:b` | $\Leftrightarrow$ | `lc __ rc` | `a` is realized as `b` **always and only** in the context `lc __ rc` |
| `a:b` | $/\Leftarrow$ | `lc __ rc` | `a` is **never** realized as `b` in the context `lc __ rc` |

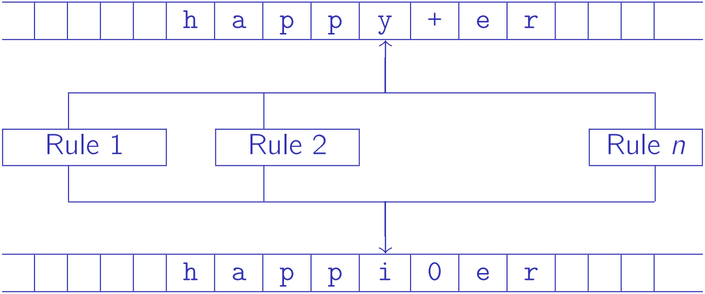
# Parallel Rules

All applicable rules are applied in parallel (provided their contexts match).



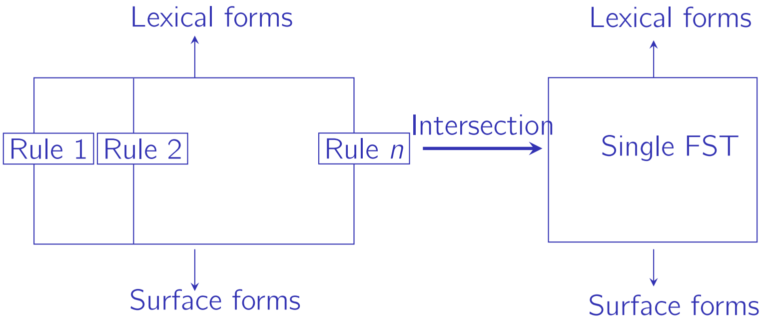
# Rules and Transducers

Each rule can be compiled into an equivalent finite-state transducer.



# Rule Intersection

The parallel transducers are subsequently combined into a single transducer by intersection.




# Problems with Intersection

The intersection of two finite-state automata is always a finite-state automaton.

This is not always true for finite-state transducers.

Kaplan and Kay (1994) showed that when the surface and lexical strings have the same length (no $\varepsilon$-transitions), the intersection is itself a transducer.

This property is sufficient for practical rule intersection.

In practice, transducers derived from two-level rules are intersected by treating the $\varepsilon$ symbol as an ordinary symbol (Beesley & Karttunen 2003, p. 55).

# Xerox

Originally the rules were compiled by hand.

This quickly becomes intractable when rules conflict or when rule contexts interact with the symbols being transduced.

The problem is solved by a compiler that automatically constructs transducers from two-level rules.

Xerox's XFST is a publicly available tool and, to date, the only robust implementation of a morphological rule compiler.

# Morphology of French Verbs

We have so far used a stem together with a set of suffixes for regular French verbs.
Irregular French verbs are considerably more complex.
Chanod (1994) provides an example of their decomposition into simple rules.

| **Infinitive** | **courir** | **dormir** | **battre** | **peindre** | **écrire** |
|---|---|---|---|---|---|
| 1st sg. | cour**s** | **dors** | **bats** | **peins** | écris |
| 2nd sg. | cour**s** | **dors** | **bats** | **peins** | écris |
| 3rd sg. | cour**t** | **dort** | **bat** | **peint** | écrit |
| 1st pl. | cour**ons** | dormons | battons | peignons | **écrivons** |
| 2nd pl. | cour**ez** | dormez | battez | peignez | **écrivez** |
| 3rd pl. | cour**ent** | dorment | battent | peignent | **écrivent** |

# French Morphology

| | | |
|---|---|---|
| **Lexical form:** stem | dormir | +IndP +SG +P1 |
| | ↕ | ↕ |
| **Intermediate form:** inflection | dorm | +IndP +SG +P1 |
| | ↕ | ↕ |
| **Intermediate form:** deletion of *m* before *s* | dorm | s |
| | ↕ | ↕ |
| **Surface form:** | dor | s |

From *peindre* to *peins*:

$$\texttt{n:0} \Leftrightarrow \texttt{g \_\_ [s|t]}$$

# Composition and Intersection